

ADVANCED PROGRAMMING ASSIGNMENT

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Program Description:

This program demonstrates different space complexities: Constant $O(1)$, Linear $O(n)$, and Quadratic $O(n^2)$. It measures actual memory usage using `sizeof()` and dynamic memory allocation.

Source Code:

```
#include <stdio.h>
#include <stdlib.h>

// Function to demonstrate O(1) Space Complexity
void constantSpace() {    int a = 5;    int b
= 10;

    printf("Constant Space Memory: %lu bytes\n", sizeof(a) + sizeof(b));
}

// Function to demonstrate O(n) Space Complexity
void linearSpace(int n) {    int *arr;

    arr = (int *)malloc(n * sizeof(int));

    printf("Linear Space Memory for n = %d: %lu bytes\n",
n, n * sizeof(int));

    free(arr);
}

// Function to demonstrate O(n^2) Space Complexity
void quadraticSpace(int n) {    int **matrix;

    matrix = (int **)malloc(n * sizeof(int *));

    for (int i = 0; i < n; i++)    {
matrix[i] = (int *)malloc(n * sizeof(int));    }

    printf("Quadratic Space Memory for n = %d: %lu bytes\n",
n,        (n * sizeof(int *)) + (n * n * sizeof(int)));

    for (int i = 0; i < n; i++)
    {        free(matrix[i]);
    }

    free(matrix);
}
```

```

int main() {
int n = 4;

constantSpace();
linearSpace(n);
quadraticSpace(n);

    return 0;
}

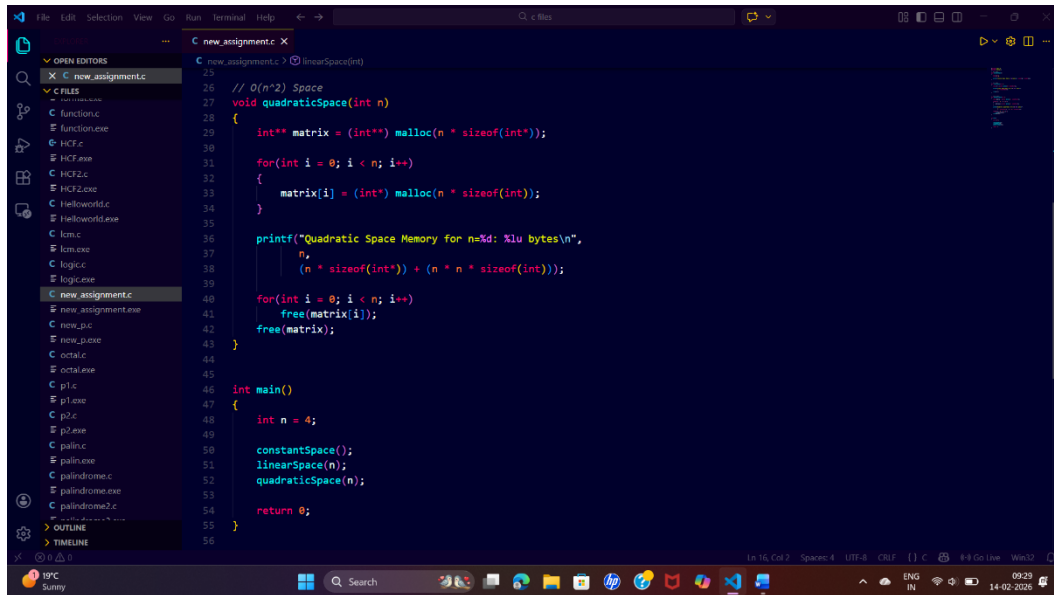
```

Screenshots of the actual program & it's outputs:

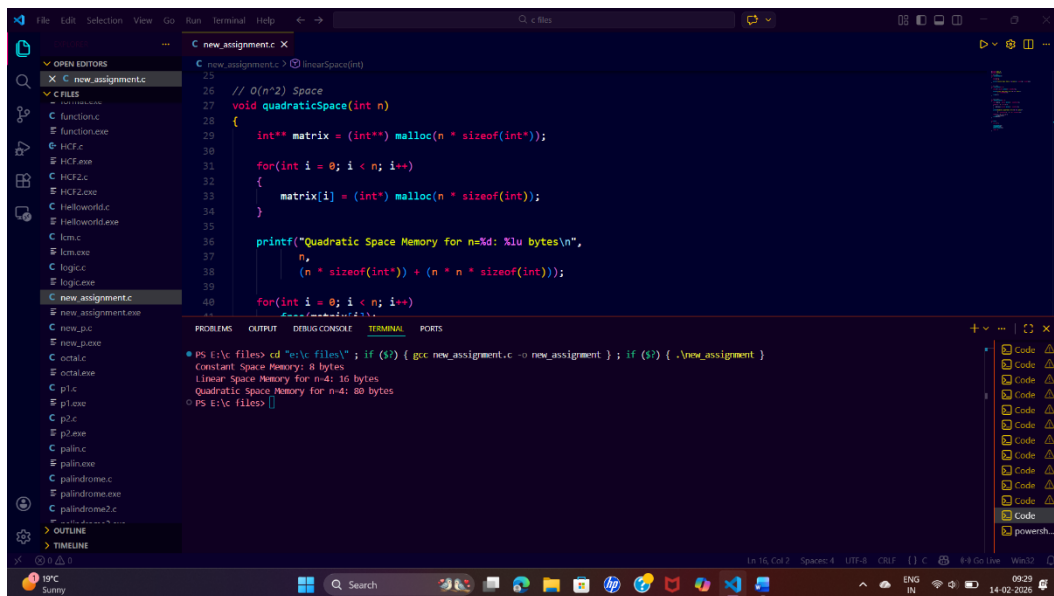
```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  // O(1) Space
5  void constantSpace()
6  {
7      int a = 5;
8      int b = 10;
9
10     printf("Constant Space Memory: %lu bytes\n", sizeof(a) + sizeof(b));
11 }
12
13 // O(n) Space
14 void linearSpace(int n)
15 {
16     int* arr = (int*) malloc(n * sizeof(int));
17
18     printf("Linear Space Memory for n=%d: %lu bytes\n",
19           n, n * sizeof(int));
20
21     free(arr);
22 }
23
24 // O(n^2) Space
25 void quadraticSpace(int n)
26 {
27     int** matrix = (int**) malloc(n * sizeof(int*));
28
29     for(int i = 0; i < n; i++)
30     {
31
32

```



```
25 // O(n^2) Space
26 void quadraticSpace(int n)
27 {
28     int** matrix = (int**) malloc(n * sizeof(int*));
29
30     for(int i = 0; i < n; i++)
31     {
32         matrix[i] = (int*) malloc(n * sizeof(int));
33     }
34
35     printf("Quadratic Space Memory for n=%d: %lu bytes\n",
36           n,
37           (n * sizeof(int*)) + (n * n * sizeof(int)));
38
39     for(int i = 0; i < n; i++)
40         free(matrix[i]);
41     free(matrix);
42 }
43
44 int main()
45 {
46     int n = 4;
47
48     constantSpace();
49     linearSpace(n);
50     quadraticSpace(n);
51
52     return 0;
53 }
```



```
PS E:\c files> cd "c:\c files\"; if ($?) { gcc new_assignment.c -o new_assignment }; if ($?) { .\new_assignment }
Constant Space Memory: 8 bytes
Linear Space Memory for n=4: 16 bytes
Quadratic Space Memory for n=4: 80 bytes
PS E:\c files>
```